

Hospital Emergency Room Dashboard

Objective: Emergency room leaders need a clear way to track wait times, patient numbers, and satisfaction scores to help the hospital run smoothly. This interactive dashboard analyzes over 9,000 patient visits to show what drives busy periods, pinpoint long wait times, and see how wait times affect patient satisfaction across different departments.

Data Source & Tools:

Primary Tool: Tableau Public

Dataset: Hospital Emergency Room Patient Records (2023–2024)

<https://www.kaggle.com/datasets/xavierberge/hospital-emergency-dataset>

Core Variables: Patient demographics (Age, Gender, Race), Department Referrals, Wait Times (minutes), and Satisfaction Scores (1–10 scale).

Key Insights Found:

- **Over 58% of ER visits resulted in no specialized department referral** (5,400 patients), while General Practice represented the highest active referral department with 1,840 patients.
- Patients aged **60+ formed the largest age group** visiting the ER, highlighting a high demand for geriatric care resources.
- Average wait times remained consistent across the monitored period, hovering between **34.1 and 36.7 minutes**
- Across the two years, the average satisfaction score ranged from **4.6 to 5.3 out of 10**.
- The ER served a diverse population, with **African American and White patients** making up the two largest racial demographics visiting the ER

Interactive Features & Technical Highlights:

- **Interactive Dashboard Actions:** Built click-to-filter controls across the dashboard. Clicking the Male or Female KPI cards, or selecting specific bars on the Age, Race, and Department Referral charts, instantly filters all other visuals to match that selection.
- **Calculated Fields & Metrics:** Created custom formulas to calculate overall metrics, including total patient count (9,216), gender breakdown (4,705 Male & 4,511 Female), and average wait times across visits.
- Added a line chart with the distinct count of Patient ID numbers for every month.